Jodh Singh Nahal

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Education

University of British Columbia (UBC) *Vancouver, BC*
Bachelor of Applied Science in Computer Engineering *Expected May 2028*
- Presidential Scholars Award ($60,000), Dean's Honour List, available 4 or 8 months, CGPA: 84.7%.

Technical Skills

HDL & Design: Verilog/SystemVerilog, RTL Design, FSM Design, FPGA, Quartus
Verification: UVM, SVA, Coverage, Synopsys VCS/Verdi, ModelSim, Waveform Analysis
Programming: C/C++, Python, Perl, Shell, ARM Assembly
Tools: Linux, Git, Docker, Jenkins, GDB, JTAG, Logic Analyzers, Oscilloscopes, LTspice, KiCad

Experience

Hardware Verification Engineering Intern *Vancouver, BC*
AMD *Sept 2026 – Present*
- Verifying the Input–Output Memory Management Unit L1 cache behavior for the AMD IOHUB architecture using UVM constrained-random and directed testbenches.
- Added stimulus, checkers, coverpoints, and SystemVerilog Assertions (SVA) for PCIe ATS.
- Debugging RTL behavior through simulation, waveform analysis, and regression testing in Synopsys VCS/Verdi.
- Expanded the constrained-random testbench to cover corner cases and improve functional coverage.

Software Test / Embedded Development Co-op *Burnaby, BC*
Xantrex *May 2024 – Aug 2025*
- Developed object-oriented, multi-threaded C++ firmware on Linux-based Raspberry Pi gateways, achieving a 7× increase in device capacity and 90% reduction in CPU usage over the prior Python implementation.
- Wrote Python scripts to autogenerate scalable C++ object-oriented code from SQLite and Microsoft Access/Excel data, eliminating manual boilerplate.
- Developed Hardware-in-the-Loop (HIL) CAN and UART simulators in Python/C++ to execute 100+ tests for embedded gateway systems, validating edge cases and stress-testing real-time firmware performance.
- Profiled the C++ firmware to eliminate performance bottlenecks.

Embedded Systems Developer *Vancouver, BC*
UBC Rover *Sept 2023 – July 2026*
- Led a team of 3 engineers to develop a distributed embedded control system integrating ESP32 (FreeRTOS), STM32, Arduino, and TI LaunchPad MCUs over SPI/I2C with aggregated data streamed over UART.
- Developed a 16-channel telemetry system on a TI C2000 (LAUNCHXL-F28069M) using ePWM-triggered ADC interrupts, streaming GPIO/ADC data over SCI/UART.
- Debugged C2000 firmware over JTAG in CCSTUDIO.
- Developed Zephyr RTOS firmware on a Teensy 4.1 for stepper-based control of a 6-axis robotic arm.

Undergraduate Research Assistant *Vancouver, BC*
UBC ECE — Supervisor: Dr. Lutz Lampe *May 2026 – Sept 2026*
- Simulated and optimized piezoelectric energy harvesting circuits in LTspice and Simulink/Simscape.
- Designed power-management PCB hardware in KiCad integrating AC–DC rectifiers and switching circuitry.
- Researching battery-less platforms for the Internet of Underwater Things via simulation of piezoelectric interface circuits.

Projects

ARC4 Hardware Decryption | *SystemVerilog, FPGA, Quartus, ModelSim*
- Designed a complete ARC4 stream cipher decryption engine in SystemVerilog with KSA and PRGA implemented as FSMs; parallelized to 8 decryption cores with coordinated on-chip memory arbitration.
- Developed verification testbenches with waveform analysis and `.memh`-based memory initialization to validate RTL functionality and timing behavior on a DE1-SoC FPGA.

Simple RISC Machine | *Verilog, FPGA, Quartus, ModelSim*
- Designed and verified a RISC CPU in Verilog comprising an ALU, register file, RAM, program counter, and FSM-based control unit executing a reduced ARM Instruction Set Architecture (ISA); validated on a FPGA.